Combustion:

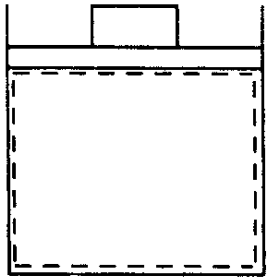
Zero-Dimensional Reactors

Instructor: Zhuyin Ren

Email: zhuyinren@tsinghua.edu.cn

Typical Zero-D Reactors

(a)
Constant-pressure fixed-mass reactor



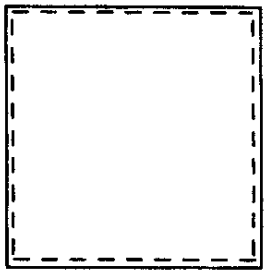
$T = T(t)$

$[X_i] = [X_i](t)$

$V = V(t)$

Perfectly-mixed, homogeneous reaction

(b)
Constant-volume fixed-mass reactor



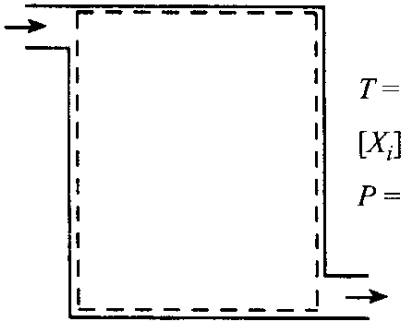
$T = T(t)$

$[X_i] = [X_i](t)$

$P = P(t)$

Perfectly-mixed, homogeneous reaction

(c)
Well-stirred reactor



T = constant

$[X_i]$ = constant

P = constant

Steady-state, steady-flow, perfectly mixed

(d)
Plug-flow reactor



$T = T(x)$

$[X_i] = [X_i](x)$

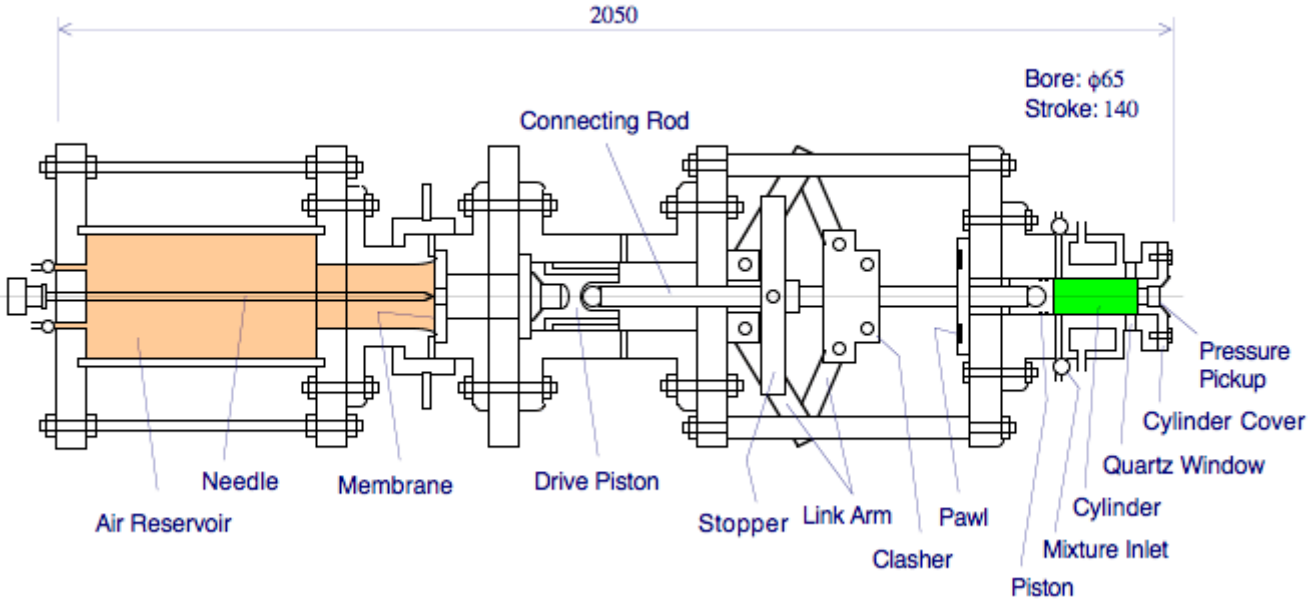
$P = P(x)$

$\mathbf{V} = v_x(x)$

Steady-state, steady-flow, no axial mixing

Examples of Constant Volume Reactors

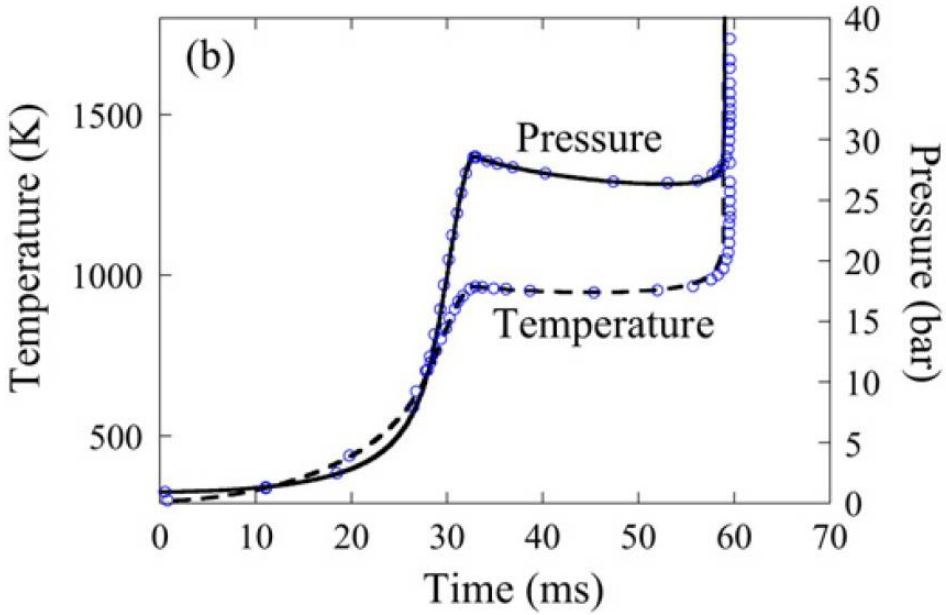
- Rapid compression machine (RCM)



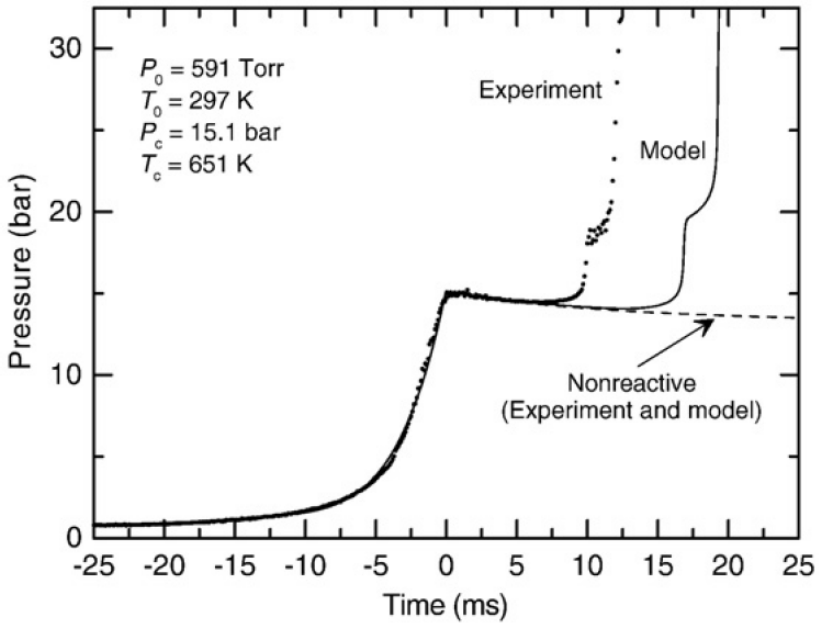


Pressure History in RCM

H2/air with one-stage ignition,
Similar behavior for small hydrocarbons
(CH4, C2H4 etc)



DME/air with two-stage ignition
Similar behavior for large hydrocarbons
(n-heptane, iso-octane etc)

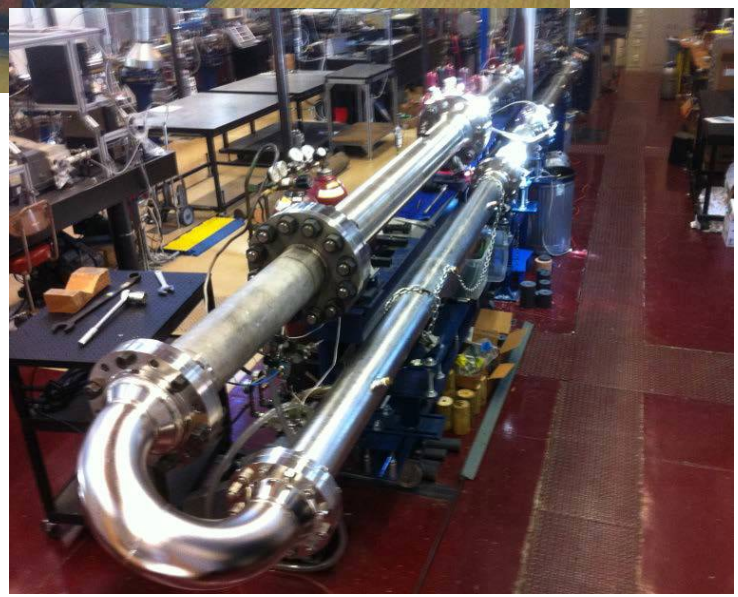


(Results from Prof. C.J. Sung at UCONN)

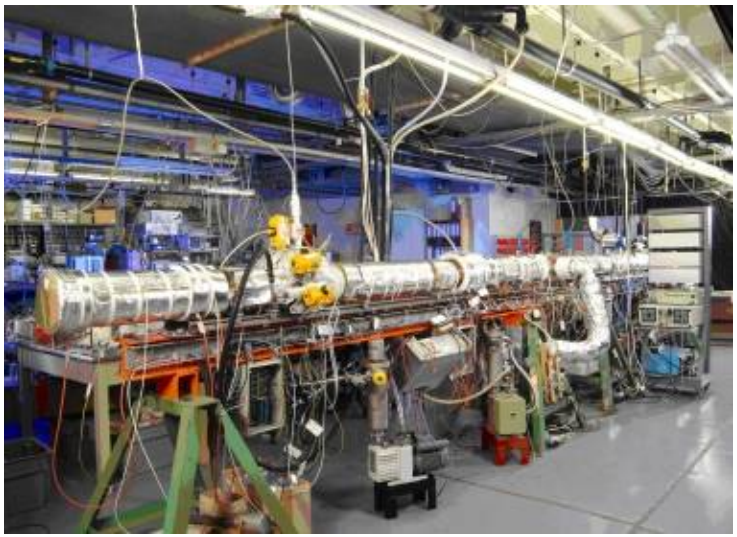
Shock Tube

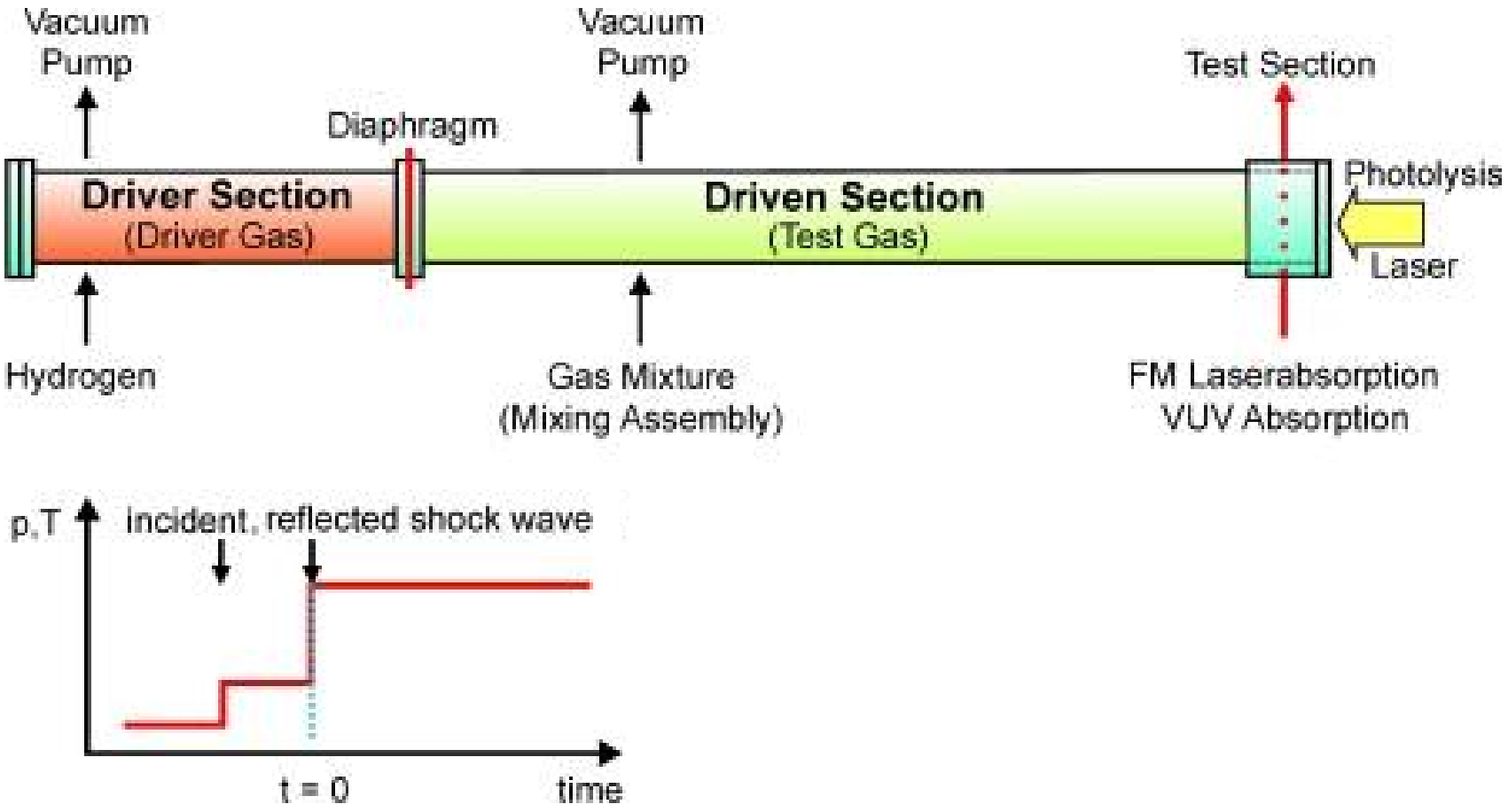
Stanford shock tube





DLR shock tube with preheating





http://www.uni-kiel.de/phc/ags/friedrichs/stossrohr_bunt_web.jpg

Normal Shock Relations

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}\left(M_1^2 - 1\right)$$

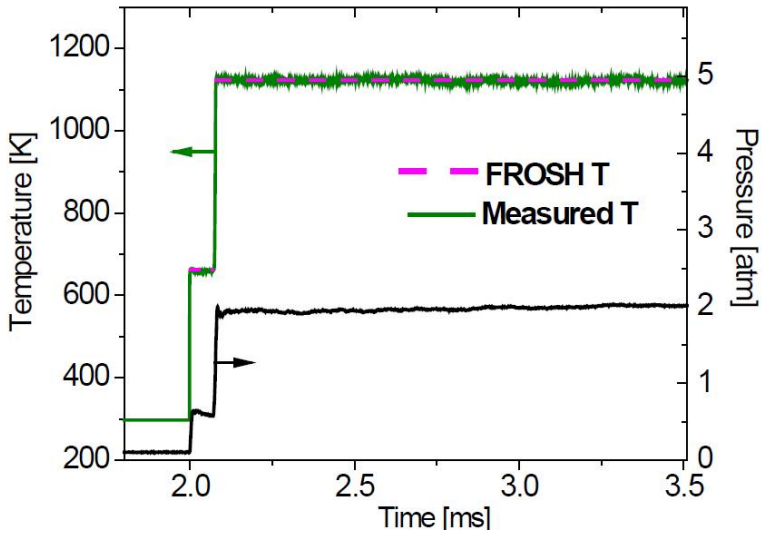
$$\frac{\rho_2}{\rho_1} = \frac{M_1^2}{1 + \frac{\gamma-1}{\gamma+1}\left(M_1^2 - 1\right)}$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma+1}\left(M_1^2 - 1\right)\right]\frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2}$$

$$M_2^2 = \frac{1 + \left(\frac{\gamma-1}{2}\right)M_1^2}{\gamma M_1^2 - \left(\frac{\gamma-1}{2}\right)}$$

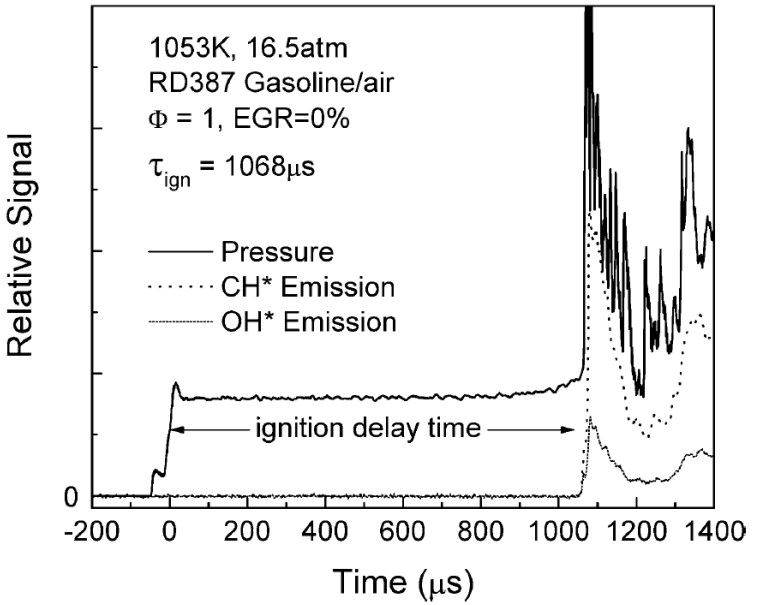
T/P History in Shock Tubes

- Non-reacting (O2/Ar)



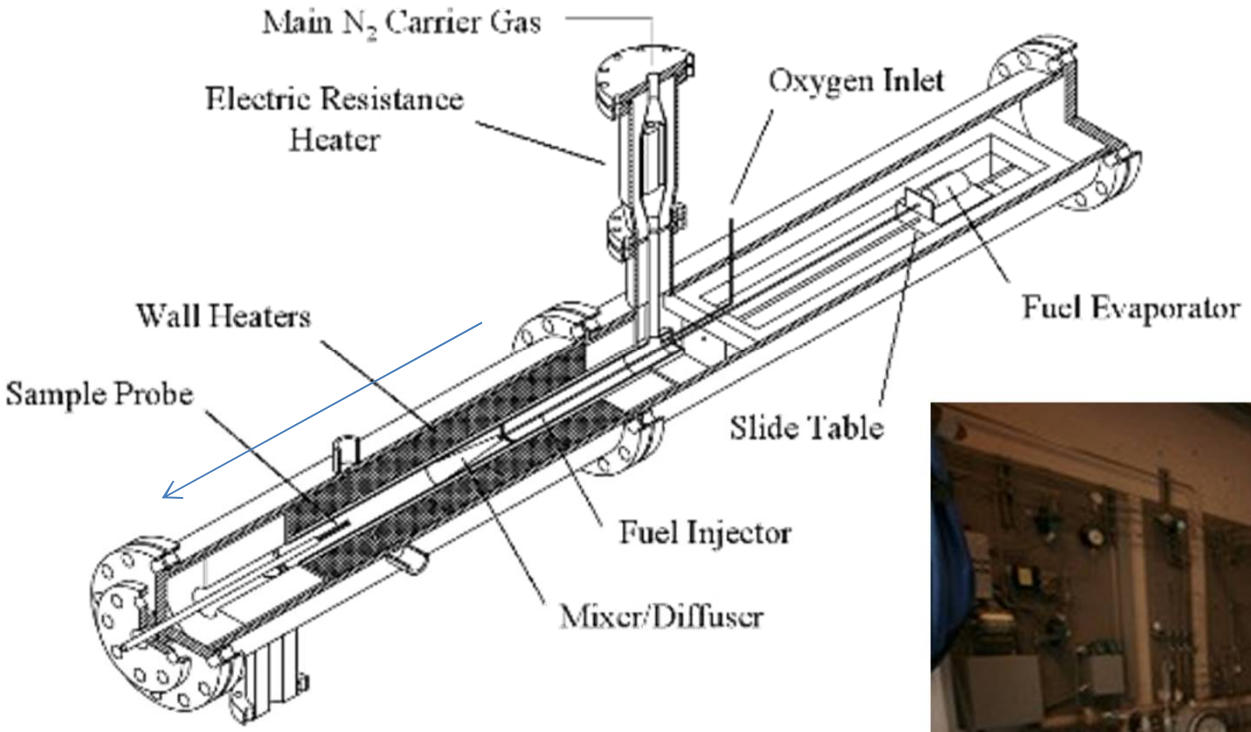
(Spearrin, Ren, Jeffries, Hanson, *Applied Physics B*, 2014)

- Reacting ixtures



(Gauthier et al. CNF 139 (2004) 300–311)

Examples of Constant-Pressure Reactors: Flow Reactor



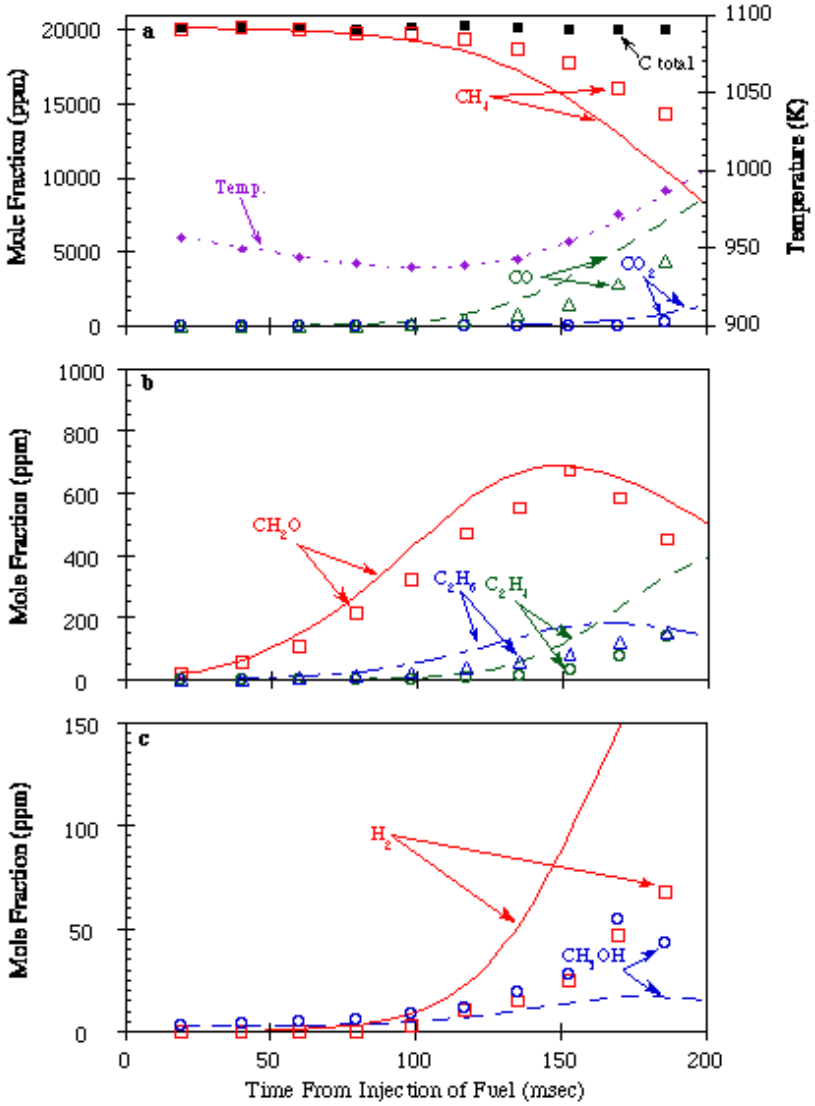
The Princeton flow reactor



Flow Reactor Species History

methane/air in a flow reactor

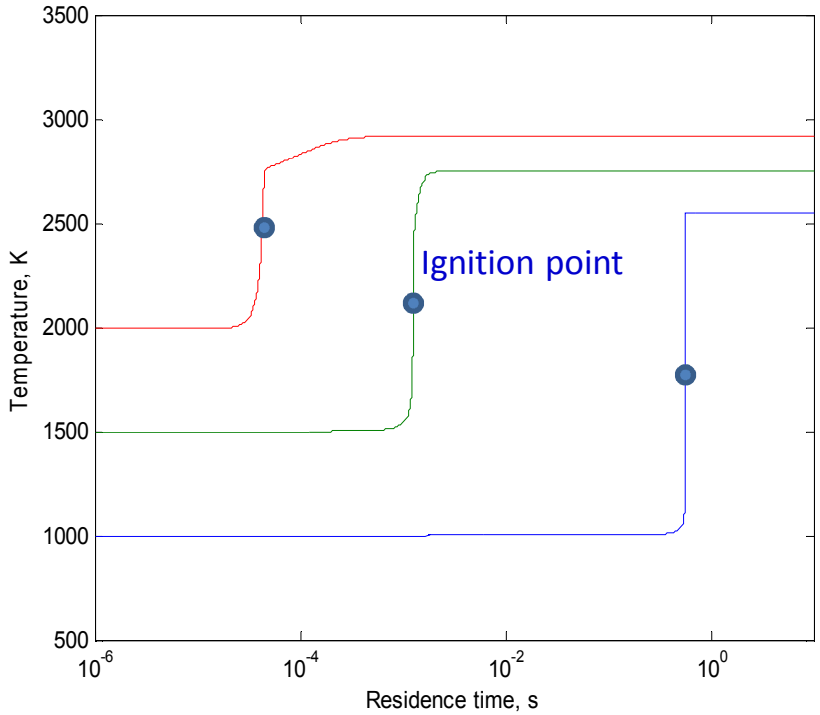
Symbols: experiment
Lines: constant P simulation
with GRI 1.2



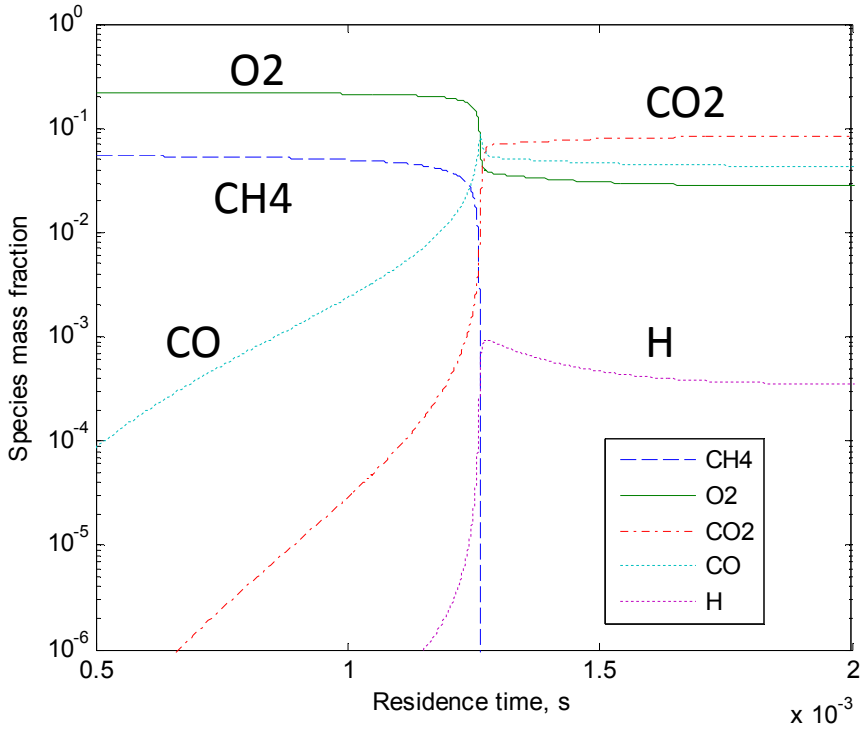
(Hunter, PennState 1994)

Auto-ignition: Small Hydrocarbons

Methane-air, ϕ = 1.0, p = 1 atm

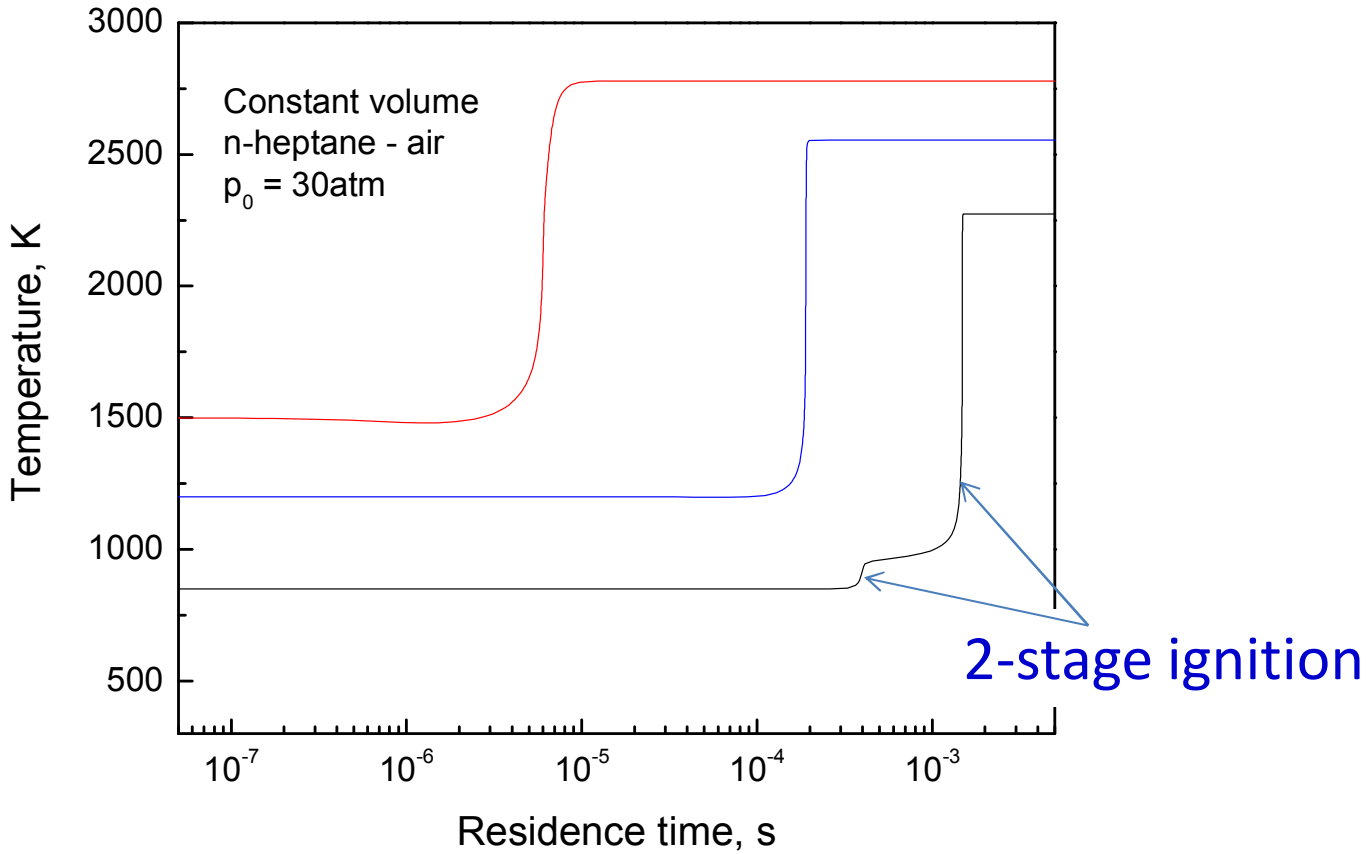


Methane-air,
ϕ = 1.0, p = 1 atm, T0 = 1500K

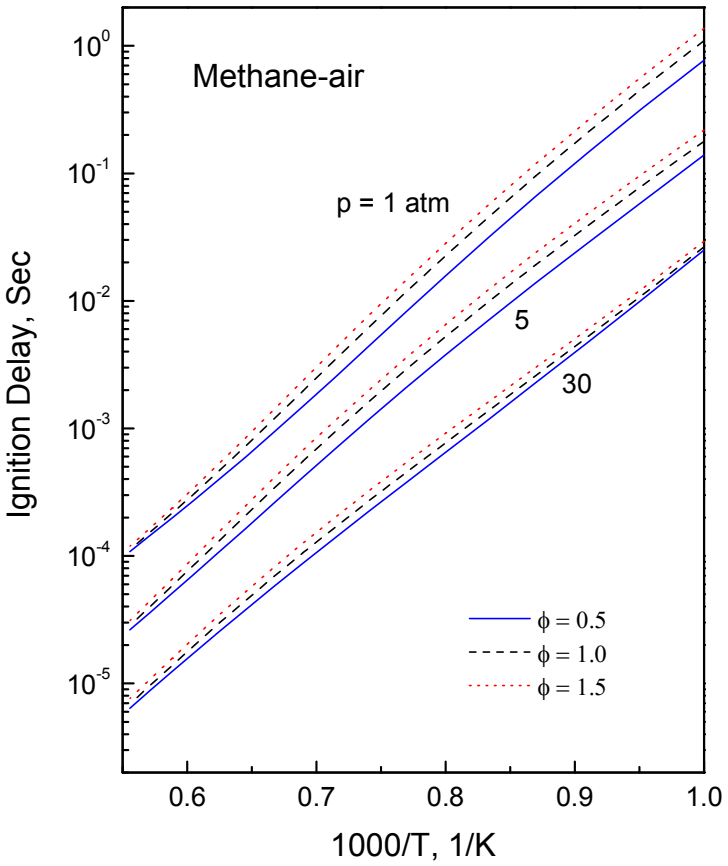


Ignition delay:
the residence time at the ignition point

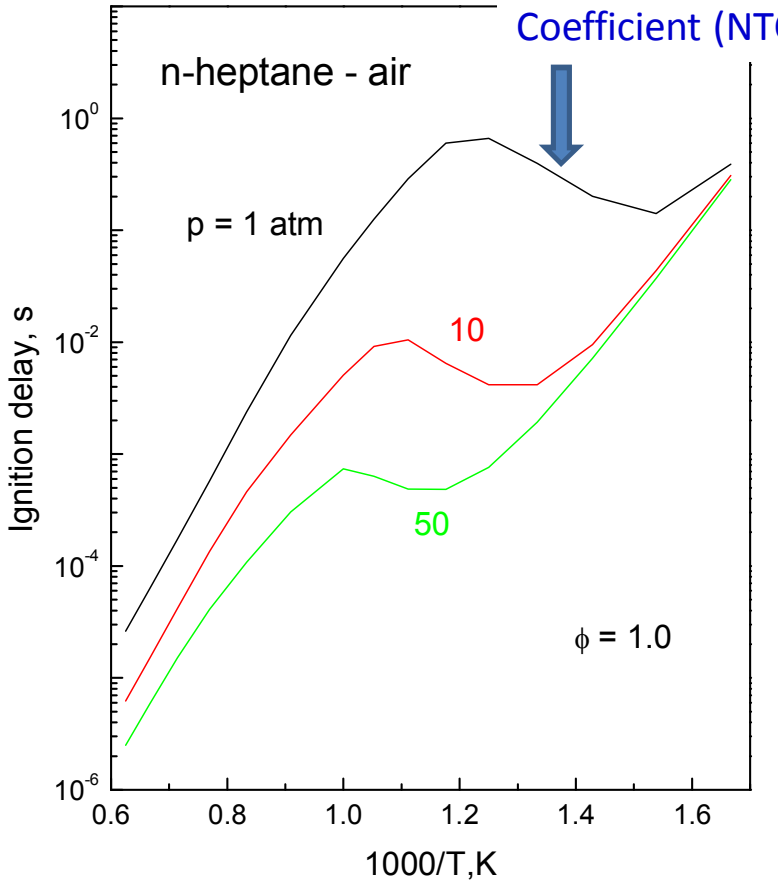
Auto-ignition:
Temperature Profiles for Large Hydrocarbons



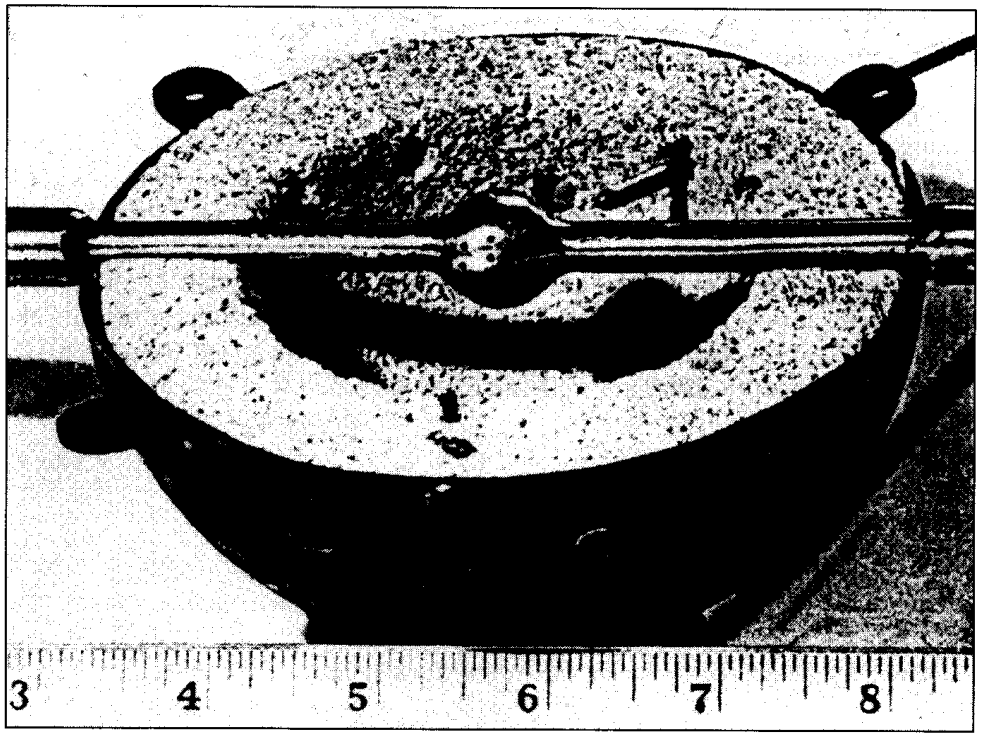
Ignition Delay:
Low vs. High Hydrocarbons

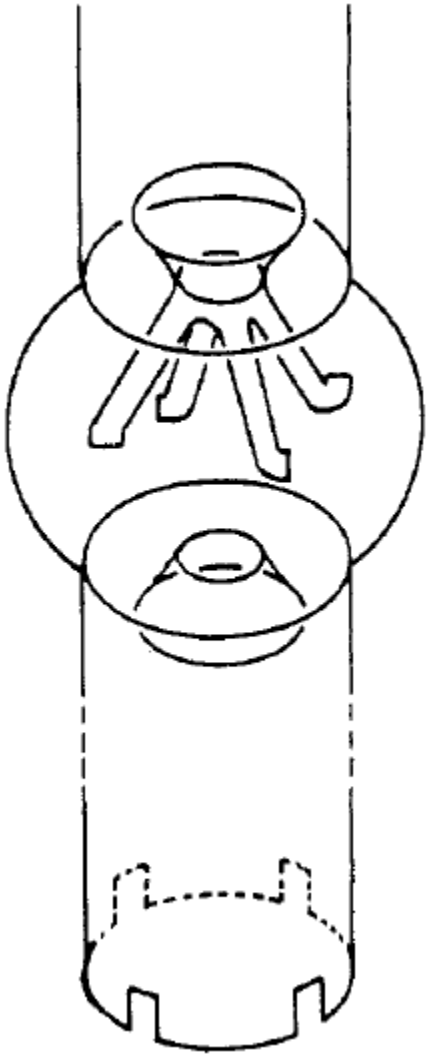


Negative Temperature Coefficient (NTC)



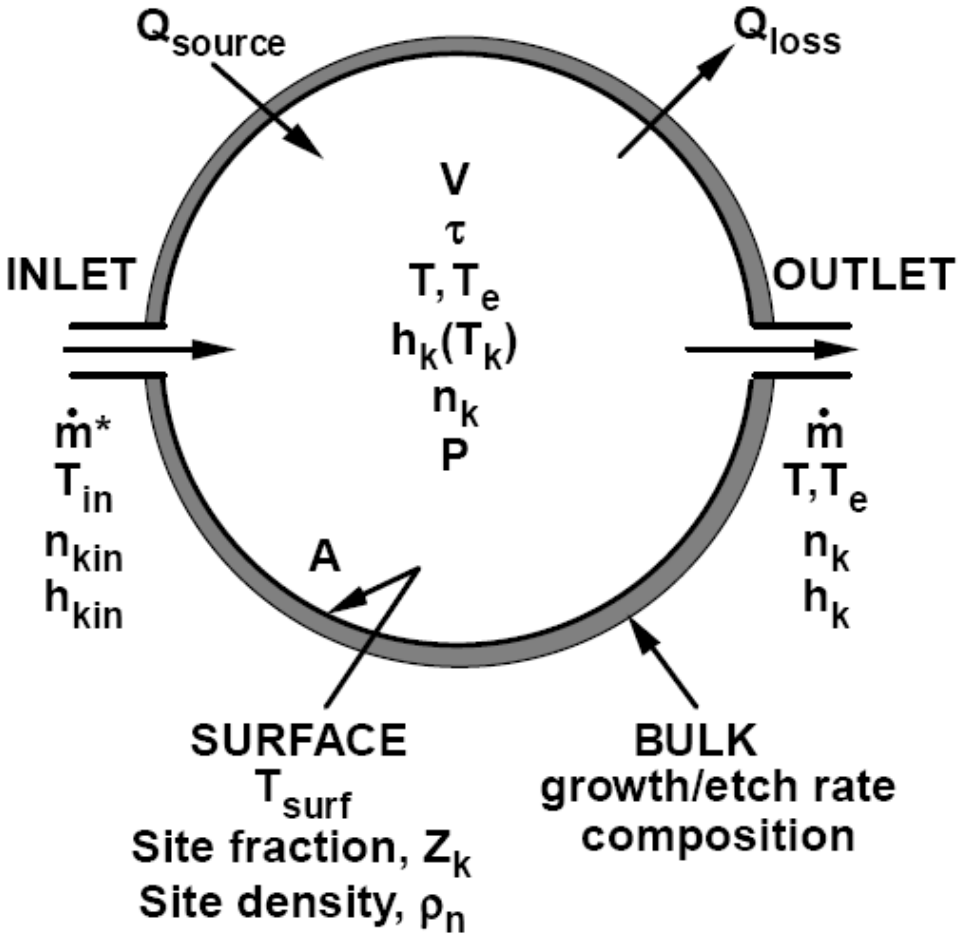
Examples of Constant Pressure Reactors: Perfectly/Well/Jet Stirred Reactors



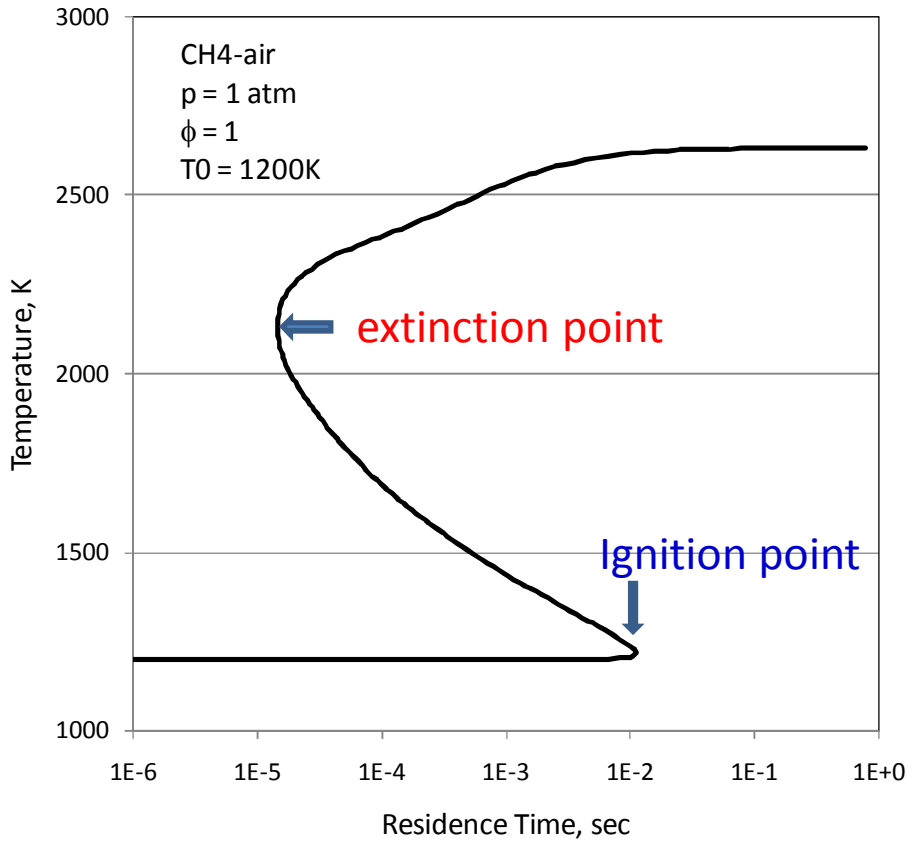


JSR

Schematics of PSR

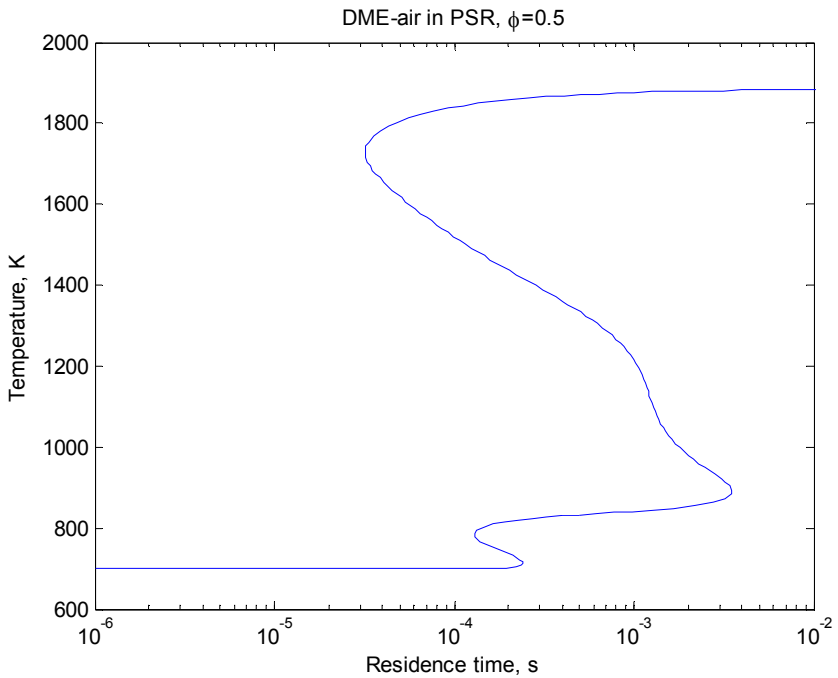


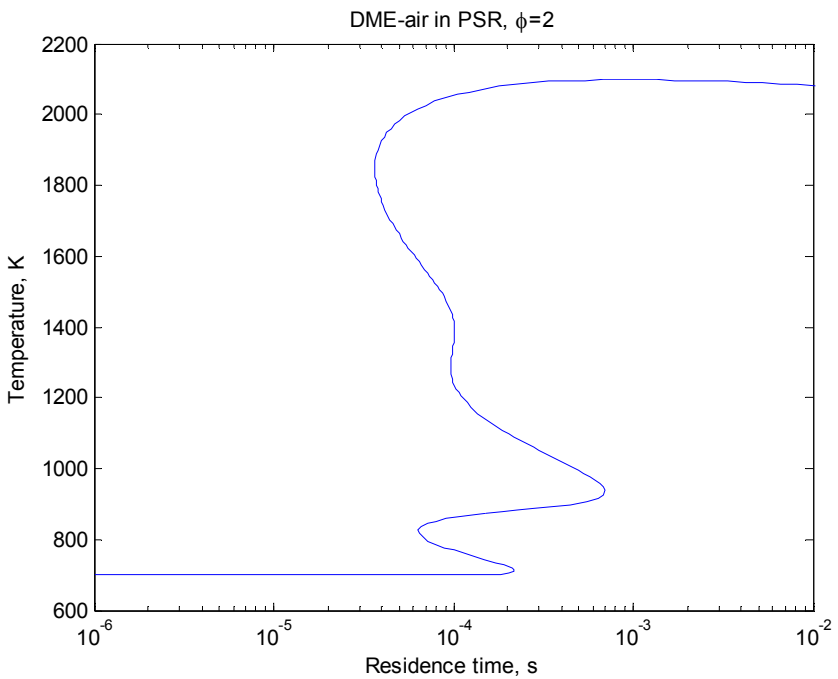
S-Curve of PSR: Steady State Combustion



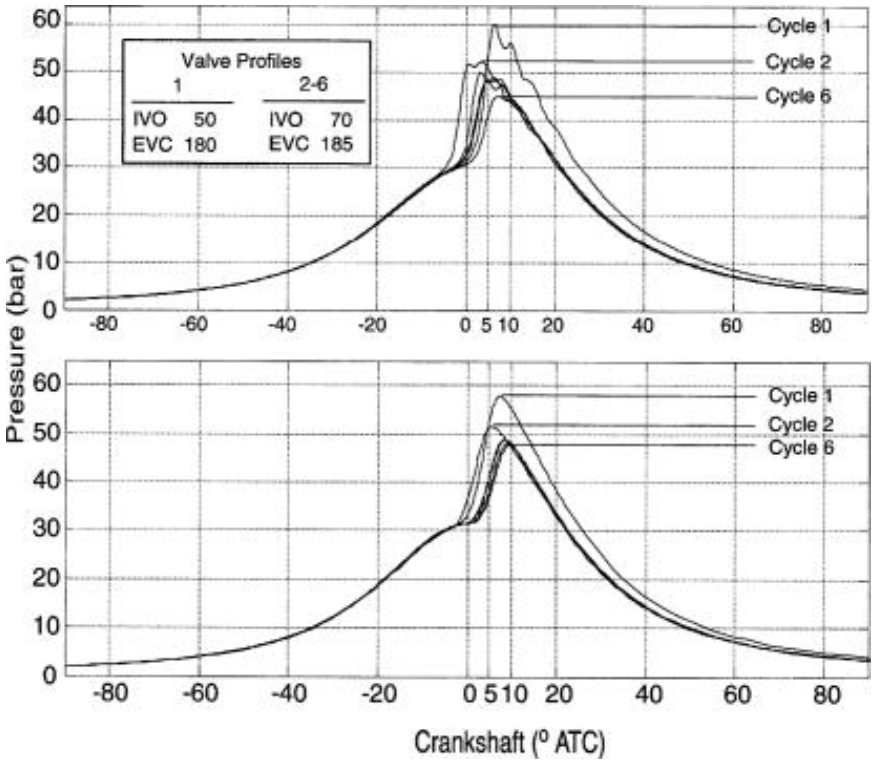
S-curve with Multiple Turnings

DME-air, p = 30 atm, T_{in}=700K

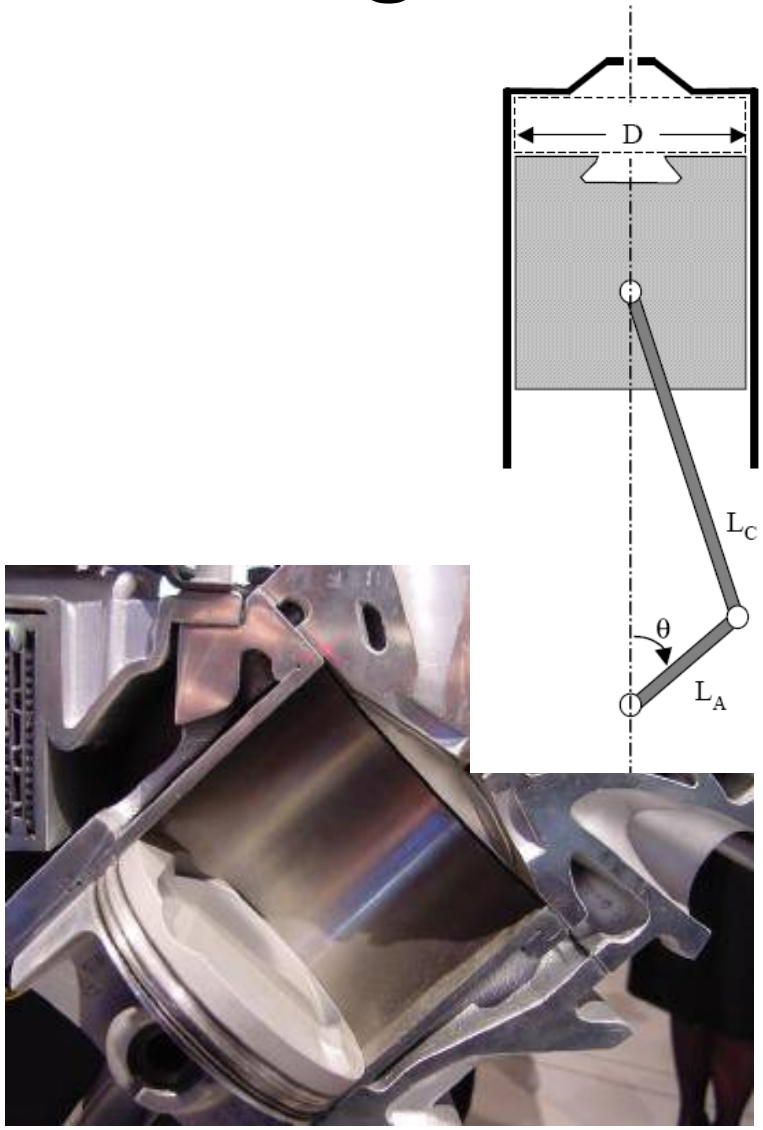




Internal Combustion Engines



(Shaver et al, 2004)



Detonation Waves: the ZND-Structure

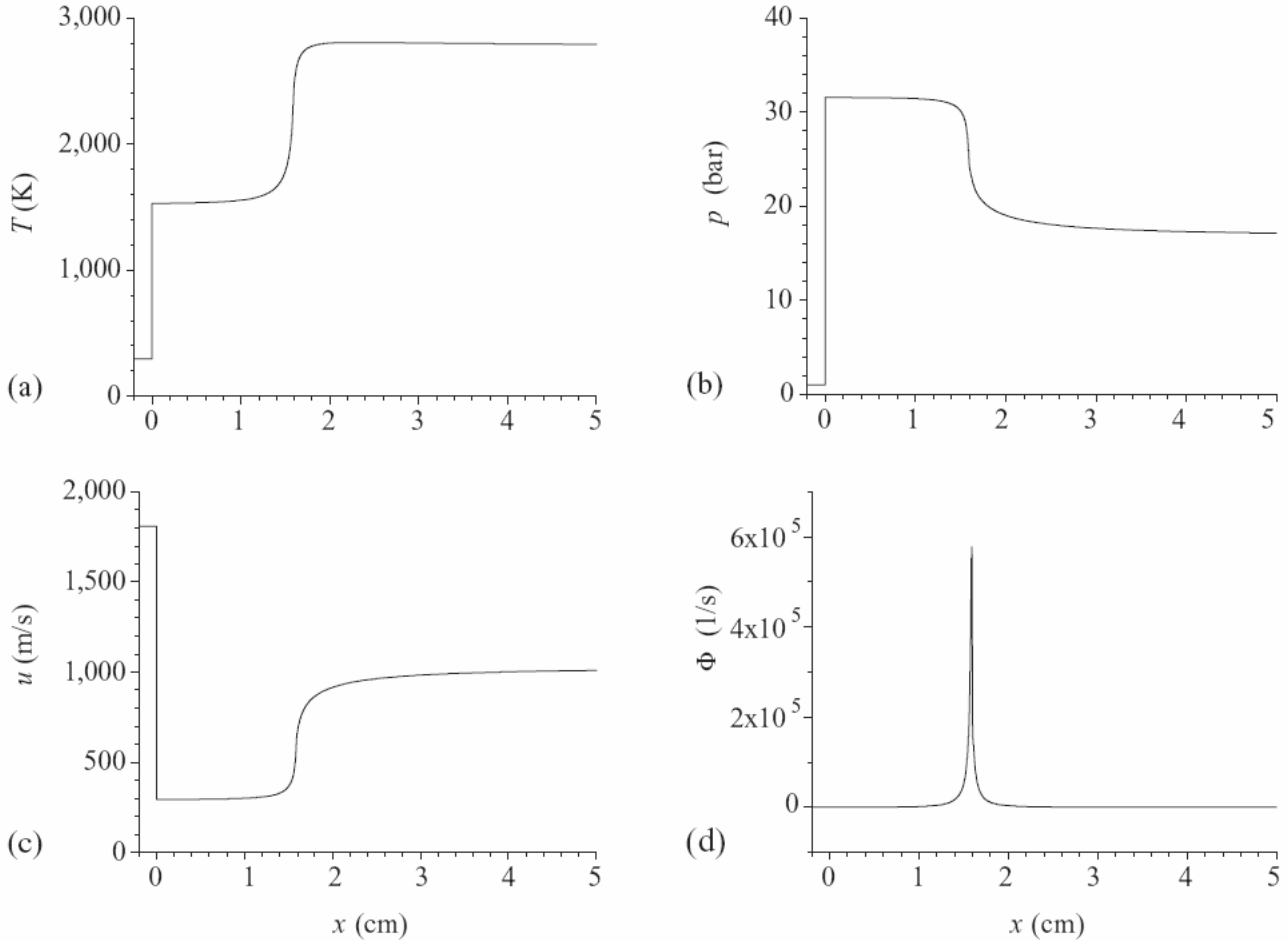
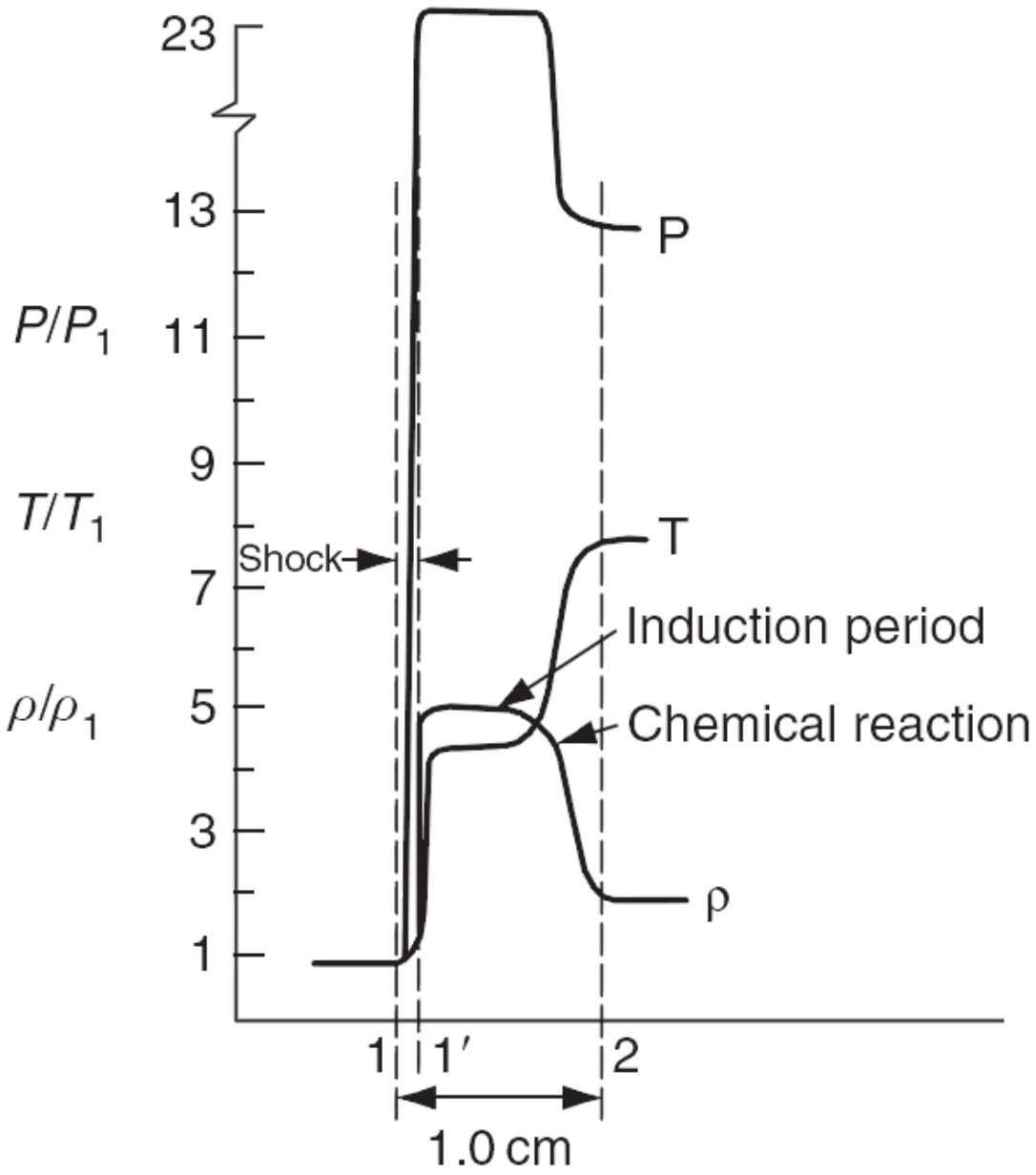


Figure 14.5.3. Calculated ZND structure of a CJ detonation in a stoichiometric methane–air mixture initially at atmospheric conditions: (a) temperature, (b) pressure, (c) gas velocity and (d) rate of energy release are plotted in terms of distance behind the leading shock.


23
13
11
9
7
5
3
1
P/P_1
T/T_1
ρ/ρ_1
Shock
P
T
ρ
Induction period
Chemical reaction
1
1′
2
1.0 cm